Exercise 3.8

Line fitting

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
data = np.loadtxt('../data/millikan.txt')
```

In [5]:
```
data = data.T
```

In [6]:
```
data
```

Out[6]:
```
array([[5.48740e+14, 6.93100e+14, 7.43070e+14, 8.21930e+14, 9.60740e+14,
        1.18400e+15],
       [5.30900e-01, 1.08420e+00, 1.27340e+00, 1.65980e+00, 2.19856e+00,
        3.10891e+00]])
```

In [4]:
```
df = pd.read_csv('../data/millikan.txt', sep=' ', names=['x', 'y'])
```

In [7]:
```
df
```

Out[7]:

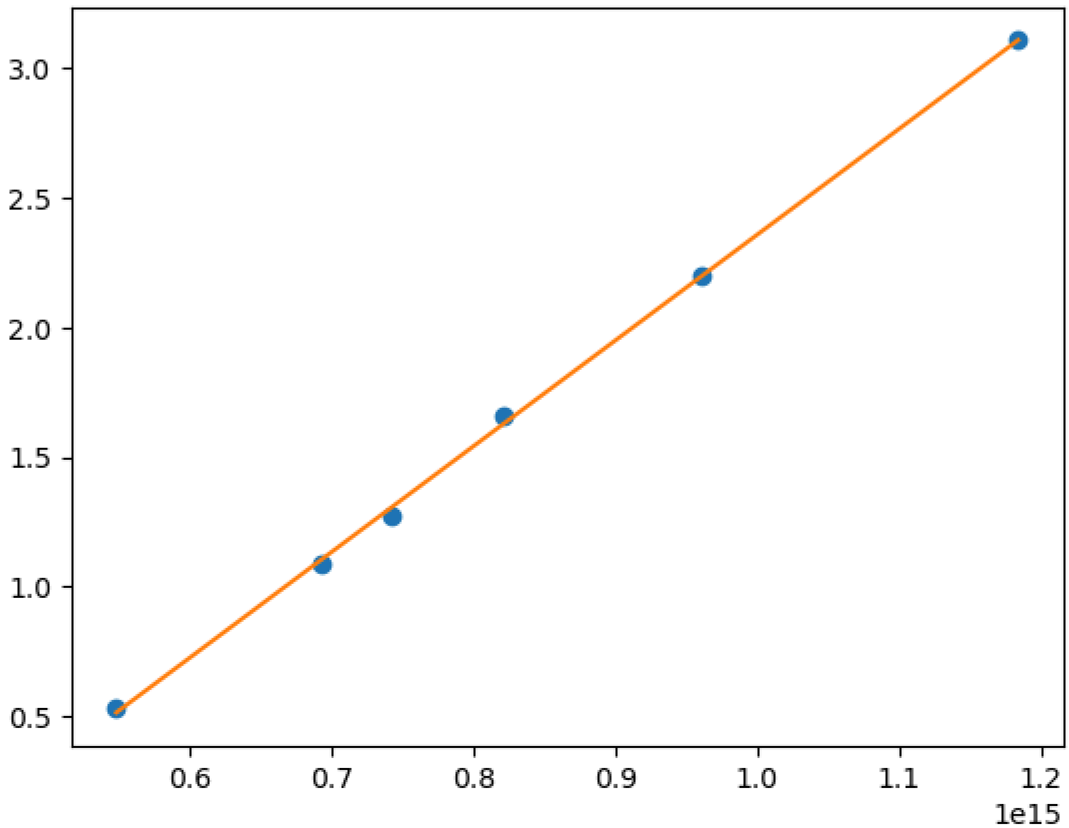
	x	y
0	5.487400e+14	0.53090
1	6.931000e+14	1.08420
2	7.430700e+14	1.27340
3	8.219300e+14	1.65980
4	9.607400e+14	2.19856
5	1.184000e+15	3.10891

In [21]:
```
n = len(df)

ex = 1/n*np.sum(df['x'])
ey = 1/n*np.sum(df['y'])
exx = 1/n*np.sum(df['x']**2)
exy  = 1/n*np.sum(df['x']*df['y'])

m = (exy - ex*ey)/(exx - ex**2)
c = (exx*ey - ex*exy)/(exx - ex**2)

print('slope = ', m)
print('y intercept = ', c)

df['fit'] = m*df['x'] + c
```

```python
fig0, ax0 = plt.subplots()

ax0.plot(df['x'], df['y'], 'o')
ax0.plot(df['x'], df['fit'])
```

```
slope =  4.088227358517502e-15
y intercept =  -1.73123580398135
```

Out[21]:
```
[<matplotlib.lines.Line2D at 0x70f8f85de1e0>]
```



In [20]:
```python
m*1.602*10**(-19)
```

Out[20]:
```
6.549340228345038e-34
```

In [ ]:
```python

```